

Notes: Ross (JF, 2015)

The Recovery Theorem

Contents

1	Big picture	2
2	Data and measurement: state prices, options, and recovered distributions	3
2.1	State prices and risk-neutral probabilities	3
2.2	Transition state prices	3
2.3	Transition-independent kernel	3
2.4	Empirical option data	3
3	Models and theoretical specifications	4
3.1	Discrete state-space recovery	4
3.2	Perron-Frobenius step	4
3.3	Kernel and natural probabilities	4
3.4	The Recovery Theorem	5
3.5	Binomial and multinomial recovery	5
3.6	Efficient-market test	6
4	Identification and empirical design: what makes recovery credible	7
4.1	What identifies recovery	7
4.2	Why irreducibility matters	7
4.3	Why transition independence matters	7
4.4	Why the empirical exercise is difficult	7
4.5	What the paper cannot fully prove	7
5	Tables and figures	9
5.1	Table I: Fixed lognormal payoff and CRRA pricing kernel	9
5.2	Figure 1 and Figure 2: option-implied state prices and recovered density	10
5.3	Table II: State prices and recovered transition price matrix	11
5.4	Table III: Recovered pricing kernel and natural transition matrix	12
5.5	Table IV: Recovered and bootstrapped natural marginal distributions	13
5.6	Table V: Recovered versus historical tail probabilities	14
6	Short summary	16
7	Conclusion	16
7.1	Core conclusion	16
7.2	Economic intuition	16
7.3	Takeaway	16

1 Big picture

- **Question.** Can we recover the market's natural, or subjective, probability distribution of future returns and the pricing kernel using state prices alone?
- **Standard obstacle.** Option prices reveal risk-neutral probabilities or state prices. But state prices equal natural probabilities times the pricing kernel. Without more structure, this product cannot be separated into its two components.
- **Core contribution.** Ross proves that under no arbitrage, an irreducible state-price transition matrix, and a transition-independent pricing kernel, the natural transition probabilities, discount factor, and pricing kernel are uniquely pinned down by the matrix of state prices.
- **Key theorem.** In discrete form, the state-price matrix P has a Perron-Frobenius eigenvalue and positive eigenvector. The eigenvalue gives the subjective discount factor, and the eigenvector gives the pricing kernel.
- **Economic intuition.** Recovery uses dynamic consistency. State prices across transitions cannot be arbitrary. The way one-period state prices compound over states places enough restrictions to separate risk aversion from beliefs.
- **Important assumption.** The pricing kernel is *transition independent*: it depends on the ending state and on the beginning state only through normalization. This is true in representative-agent models with time-separable utility, but it is a strong restriction.
- **Empirical illustration.** Ross estimates state price densities from S&P 500 option prices, constructs an approximate state-price transition matrix, and applies recovery.
- **Main empirical message.** The recovered natural distribution has a fatter left tail than the historical bootstrap distribution. Option prices imply a higher market-perceived probability of large crashes than historical data alone suggests.
- **Risk premium implication.** The recovered kernel can be used to infer an expected return, risk premium, Sharpe-ratio bound, and a model-free type of efficient-market test.
- **Caution.** Ross emphasizes that the empirical application is illustrative. Estimating the full transition state-price matrix from option data is hard, and small estimation choices can materially affect recovered beliefs.

2 Data and measurement: state prices, options, and recovered distributions

2.1 State prices and risk-neutral probabilities

The basic pricing object is the Arrow-Debreu state price $p(\theta)$, the current price of a claim that pays one dollar in state θ and zero otherwise. For a payoff $g(\theta)$,

$$p_g = \int g(\theta) dP(\theta). \quad (1)$$

The same price can be written as a discounted risk-neutral expectation or as a natural expectation weighted by the pricing kernel:

$$p_g = e^{-r(\theta_0)T} E^*[g(\theta)] = E[g(\theta)\phi(\theta)]. \quad (2)$$

Interpretation. State prices are observable from derivative prices in principle. Natural probabilities and the pricing kernel are not separately observable without additional structure.

2.2 Transition state prices

Ross works with transition state prices. Let θ_i be the current state and θ_j be a future state. The one-period state price is

$$P(\theta_i, \theta_j) = e^{-r(\theta_i)} Q(\theta_i, \theta_j), \quad (3)$$

where Q is the risk-neutral transition probability. The pricing kernel is

$$\phi(\theta_i, \theta_j) = \frac{p(\theta_i, \theta_j)}{f(\theta_i, \theta_j)}. \quad (4)$$

Interpretation. Recovery is about the transition object P , not just a single terminal density. Knowing state prices at multiple horizons is what creates the dynamic restrictions.

2.3 Transition-independent kernel

The central identifying assumption is

$$\phi(\theta_i, \theta_j) = \delta \frac{h(\theta_j)}{h(\theta_i)}, \quad (5)$$

where $h(\theta) > 0$ and $\delta > 0$. In a representative-agent model with additively separable utility,

$$\phi(\theta_i, \theta_j) = \delta \frac{U'(c(\theta_j))}{U'(c(\theta_i))}. \quad (6)$$

Interpretation. The kernel is path independent: only the initial and terminal states matter. This is the main restriction that makes recovery possible.

2.4 Empirical option data

- The empirical section uses S&P 500 option prices from a bank over-the-counter bid/offer sheet.
- Ross estimates state price densities across maturities and strike prices from option-implied volatility data.
- The state grid is chosen as stock-return ranges from -5 to $+5$ standard deviations from the current level.
- The empirical transition matrix is approximated from state-price vectors at different horizons using the forward equation.

- The historical comparison uses bootstrapped monthly S&P 500 returns from 1960 to 2010.

Interpretation. The empirical exercise is deliberately illustrative. It shows what recovery can reveal, but the paper is explicit that better state-price estimation is needed for applied use.

3 Models and theoretical specifications

3.1 Discrete state-space recovery

In the finite-state version, the state-price equations can be written as

$$DP = \delta FD, \quad (7)$$

where

- P is the matrix of Arrow-Debreu state prices p_{ij} ;
- F is the natural transition matrix f_{ij} ;
- D is a diagonal matrix containing the undiscounted pricing kernel;
- δ is the subjective discount factor.

Solving for F gives

$$F = \frac{1}{\delta} DPD^{-1}. \quad (8)$$

Interpretation. If we knew the kernel matrix D , recovery would be trivial. The theorem shows that D itself can be inferred from the eigenstructure of P .

3.2 Perron-Frobenius step

Let e be the vector of ones. Since each row of F sums to one,

$$Fe = e. \quad (9)$$

Combining this with the state-price equation gives

$$Pz = \delta z, \quad (10)$$

where

$$z \equiv D^{-1}e. \quad (11)$$

Interpretation. The recovery problem becomes a characteristic-root problem. The positive eigenvalue δ and positive eigenvector z identify the discount factor and the pricing kernel.

3.3 Kernel and natural probabilities

Given the positive eigenvector z ,

$$\frac{U'(c(\theta_i))}{U'(c(\theta_1))} = \left(\frac{1}{\delta}\right) \phi_i = d_{ii} = \frac{1}{z_i}. \quad (12)$$

Then

$$F = \left(\frac{1}{\delta}\right) DPD^{-1}. \quad (13)$$

Interpretation. The natural probabilities are the risk-neutral state prices “unadjusted” by the recovered kernel.

3.4 The Recovery Theorem

... to go from i to j , say, the steps by going from i to k to j or along a path with k steps. A matrix P is irreducible if there is always some path such that any state j can be reached from any state i .³

THEOREM 1 (The Recovery Theorem): *If there is NA, if the pricing matrix is irreducible, and if it is generated by a transition independent kernel, then there exists a unique (positive) solution to the problem of finding the natural probability transition matrix, F , the discount rate, δ , and the pricing kernel, ϕ . In other words, for any given set of state prices there is a unique compatible natural measure and a unique pricing kernel.*

Proof: Existence can also be proven directly, but it follows immediately from the fact that P is assumed to be generated from F and D as shown above. The problem of solving for F is equivalent to finding the characteristic roots (eigenvalues) and characteristic vectors (eigenvectors) of P since, if we know δ and z such that

$$Pz = \delta z, \quad (22)$$

the kernel can be found from $z = D^{-1}e$.

From the Perron Frobenius Theorem (see Meyer (2000)) all nonnegative irreducible matrices have a unique positive characteristic vector, z , and an associated positive characteristic root, λ . The characteristic root $\lambda = \delta$ is the subjective rate of time discount. Letting z denote the unique positive characteristic vector with root λ , we can solve for the kernel as

$$\frac{U'(c(\theta_i))}{U'(c(\theta_1))} = \left(\frac{1}{\delta}\right) \phi_i = d_{ii} = \frac{1}{z_i}. \quad (23)$$

To obtain the natural probability distribution, from our previous analysis,

$$F = \left(\frac{1}{\delta}\right) DPD^{-1} \quad (24)$$

³ Note that since the transition matrix is stochastic, the sum of the elements of z must be 1.

Figure 1: Original theorem/equation panel: Recovery Theorem and Perron-Frobenius recovery step.

Interpretation. Under the theorem's assumptions, there is no need to bring in historical return data or impose a parametric utility function. The state-price transition matrix itself contains enough information.

3.5 Binomial and multinomial recovery

The paper also gives a binomial and multinomial recovery result. The key lesson is that full rank of the transition matrix matters. If the matrix is full rank, recovery is pinned down. If it is not full rank, the possible recovered distributions can be restricted but are not generally unique.

and

$$p_{a,1} = 2\delta^2 f(1 - f), \quad (47)$$

which, with positivity, has a one-dimensional set of solutions.

However, in the special case where the interest rate is state independent even if the matrix is of less than full rank, risk-neutrality is one of the potential solutions. We summarize these results in the following theorem.

THEOREM 4: *(The Multinomial Recovery Theorem): Under the assumed conditions on the process and the kernel, the transition probability matrix and the subjective rate of discount of a binomial (multinomial) process can be recovered at each node from a full rank state price transition matrix alone. If the transition matrix is of less than full rank, then we can restrict the potential solutions, but recovery is not unique. If the state prices are independent of the state, then risk-neutrality is always one possible solution.*

Proof: See the analysis above preceding the statement of the theorem.

Q.E.D.

Figure 2: Original theorem panel: Multinomial Recovery Theorem.

Interpretation. This theorem clarifies why some binomial or recombining structures can be underidentified. Recovery is strongest when the state-price transition matrix is rich enough to have a unique positive eigenstructure.

3.6 Efficient-market test

Ross uses the recovered kernel to bound the predictive power of any strategy. The Hansen-Jagannathan logic gives

$$\sigma(\phi) \geq e^{-rT} \frac{|\mu|}{\sigma}, \quad (14)$$

so the volatility of the pricing kernel is an upper bound on the Sharpe ratio. In the empirical illustration,

$$\sigma^2(\phi) = 0.1065, \quad \sigma(\phi) = 0.3264. \quad (15)$$

This implies an approximate upper bound of about 10% on the annual R^2 of a return-predictability regression based on the stock-market filtration.

Interpretation. Under the maintained assumptions, a publicly implementable return-predicting strategy with R^2 above that bound would violate market efficiency independently of a specific parametric asset-pricing model.

4 Identification and empirical design: what makes recovery credible

4.1 What identifies recovery

- The observed object is the matrix of transition state prices P .
- No arbitrage ensures positive state prices.
- Irreducibility ensures all states communicate with each other.
- Transition independence makes the kernel multiplicatively separable across starting and ending states.
- Perron-Frobenius then gives a unique positive eigenvalue and eigenvector.

Interpretation. Identification comes from dynamic structure, not from historical realized returns.

4.2 Why irreducibility matters

If the state-price matrix is reducible, some future states are not reachable from some current states. Then the positive eigenvector may not be unique, and the kernel may not be pinned down.

Interpretation. Recovery is a global state-space statement. The states must form a connected transition system.

4.3 Why transition independence matters

Without transition independence, the kernel $\phi(\theta_i, \theta_j)$ would be an unrestricted function of the full transition pair. Then $P = \phi F$ would contain more unknowns than equations.

Interpretation. This is the paper's most economically substantive assumption. It rules out kernels whose pricing of a state depends on the path in richer ways.

4.4 Why the empirical exercise is difficult

- Option prices give terminal state-price densities, not the full transition matrix directly.
- The full transition matrix must be inferred from state prices at several tenors using the forward equation.
- Option data are noisy, sparse in tails, and quoted as implied volatilities rather than state prices.
- Tail probabilities are especially sensitive to interpolation and smoothing.
- Ross notes that the quarterly recovered discount root is slightly above one, a sign of estimation sensitivity.

Interpretation. The theorem is sharp. The empirical implementation is hard. The paper is best read as opening a methodology rather than delivering final empirical estimates.

4.5 What the paper cannot fully prove

- It does not prove that transition independence is true in real markets.
- It does not solve all continuous-state recovery problems.
- It does not show that option markets provide a perfectly measured state-price transition matrix.
- The empirical application uses one option date and is meant to be illustrative.
- The recovered crash probabilities may change with better smoothing, broader option data, or alternative state discretization.

Interpretation. The theorem is conceptually powerful, but applied recovery requires careful empirical design.

5 Tables and figures

5.1 Table I: Fixed lognormal payoff and CRRA pricing kernel

Read:

- This table is a controlled example with a fixed lognormal future payoff and CRRA risk aversion $\gamma = 3$.
- Panel A reports the state-space transition matrix of Arrow-Debreu prices.
- Panel B reports the natural probability transition matrix recovered from the state-price matrix.
- In this example, the rows of the recovered natural matrix are identical and match the imposed lognormal distribution.
- The recovered kernel is strongly decreasing in stock value: roughly 20.086 in the far-left state, 1 at the current state, and 0.05 in the far-right state.

Economic message: The example shows the theorem working in a transparent setting. State prices alone recover both the natural probabilities and the kernel.

Table I
Fixed Lognormally Distributed Future Payoff and a Constant
Relative Risk Aversion($\gamma = 3$) Pricing Kernel

The matrices below are derived from the one-period model presented in Section IV. The rows and columns in the matrices refer to ranges for the stock price state variable, for example, three standard deviations from the current level is 1.82. The Sigma = 0 row is the current state.

Panel A: The State Space Transition Matrix (P)												
	Sigmas	−5	−4	−3	−2	−1	0	1	2	3	4	5
Sigmas	$S_0 \backslash S_T$	0.37	0.45	0.55	0.67	0.82	1	1.22	1.49	1.82	2.23	2.72
−5	0.37	0.000	0.000	0.001	0.005	0.015	0.019	0.008	0.001	0.000	0.000	0.000
−4	0.45	0.000	0.000	0.001	0.008	0.028	0.034	0.015	0.003	0.000	0.000	0.000
−3	0.55	0.000	0.000	0.002	0.015	0.051	0.062	0.028	0.005	0.000	0.000	0.000
−2	0.67	0.000	0.000	0.003	0.028	0.092	0.113	0.051	0.008	0.001	0.000	0.000
−1	0.82	0.000	0.000	0.006	0.051	0.168	0.205	0.092	0.015	0.001	0.000	0.000
0	1	0.000	0.000	0.010	0.092	0.306	0.374	0.168	0.028	0.002	0.000	0.000
1	1.22	0.000	0.001	0.019	0.168	0.558	0.681	0.306	0.051	0.003	0.000	0.000
2	1.49	0.000	0.001	0.034	0.306	1.016	1.241	0.558	0.092	0.006	0.000	0.000
3	1.82	0.000	0.003	0.062	0.558	1.852	2.262	1.016	0.168	0.010	0.000	0.000
4	2.23	0.000	0.005	0.113	1.016	3.374	4.121	1.852	0.306	0.019	0.000	0.000
5	2.72	0.000	0.008	0.205	1.852	6.148	7.509	3.374	0.558	0.034	0.001	0.000
Kernel, $\phi =$	20.086	11.023	6.05	3.32	1.822	1	0.549	0.301	0.165	0.091	0.05	

Panel B: The Natural Probability Transition Matrix (F)												
	Sigmas	−5	−4	−3	−2	−1	0	1	2	3	4	5
Sigmas	$S_0 \backslash S_T$	0.37	0.45	0.55	0.67	0.82	1	1.22	1.49	1.82	2.23	2.72
−5	0.37	0.000	0.000	0.002	0.028	0.171	0.381	0.312	0.094	0.010	0.000	0.000
−4	0.45	0.000	0.000	0.002	0.028	0.171	0.381	0.312	0.094	0.010	0.000	0.000
−3	0.55	0.000	0.000	0.002	0.028	0.171	0.381	0.312	0.094	0.010	0.000	0.000
−2	0.67	0.000	0.000	0.002	0.028	0.171	0.381	0.312	0.094	0.010	0.000	0.000
−1	0.82	0.000	0.000	0.002	0.028	0.171	0.381	0.312	0.094	0.010	0.000	0.000
0	1	0.000	0.000	0.002	0.028	0.171	0.381	0.312	0.094	0.010	0.000	0.000
1	1.22	0.000	0.000	0.002	0.028	0.171	0.381	0.312	0.094	0.010	0.000	0.000
2	1.49	0.000	0.000	0.002	0.028	0.171	0.381	0.312	0.094	0.010	0.000	0.000
3	1.82	0.000	0.000	0.002	0.028	0.171	0.381	0.312	0.094	0.010	0.000	0.000
4	2.23	0.000	0.000	0.002	0.028	0.171	0.381	0.312	0.094	0.010	0.000	0.000
5	2.72	0.000	0.000	0.002	0.028	0.171	0.381	0.312	0.094	0.010	0.000	0.000

Table 1: Original Table I: lognormal example and CRRA kernel.

5.2 Figure 1 and Figure 2: option-implied state prices and recovered density

Read: Figure 1:

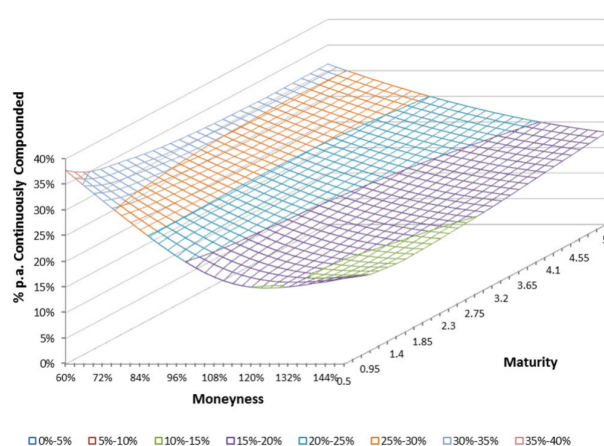
- The surface shows option-implied volatilities as a function of moneyness and maturity.
- The volatility surface is the raw market object from which the state-price densities are estimated.
- The pronounced shape of the surface reflects non-normal risk-neutral pricing, especially tail pricing.

Read: Figure 2:

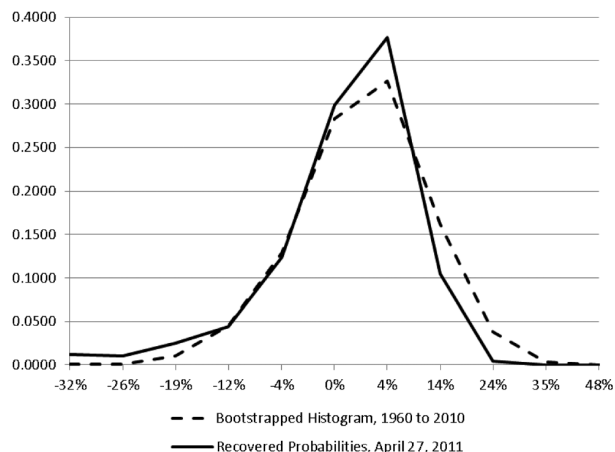
- The recovered natural density is compared with a bootstrapped historical density from S&P 500 returns.

- The recovered density has a much fatter left tail.
- It assigns more probability to large negative six-month returns than historical bootstrapping does.

Economic message: Option prices reveal tail concerns that are difficult to infer from realized returns alone. Recovery converts that option-implied information into a forward-looking natural distribution.



(a) Original Figure 1: implied volatility surface.



(b) Original Figure 2: recovered vs. bootstrapped natural densities.

5.3 Table II: State prices and recovered transition price matrix

Read:

- Panel A reports state prices for dollar payoffs in stock-return ranges at different tenors.
- The row labeled Discount Factor sums each column and gives the implied risk-free discount factor.
- Panel B estimates a transition state-price matrix consistent with the terminal state prices.
- This matrix is the input for the Recovery Theorem.

Economic message: Recovery needs transition prices, not just terminal densities. Table II is the empirical bridge from option prices to the state-price matrix.

Table II
State Prices and Recovered Probabilities

Panel A displays the Arrow-Debreu (1952) state prices for the current values of \$1 in the relevant stock price return range given in the left-hand column at the tenors given in the top row. These are derived by taking the numerical second derivative with respect to the strikes of traded call option prices from a bank offer sheet. The row labeled “discount factor” sums each column of the first state price matrix to obtain the implied risk-free discount factors. Panel B is the estimated table of contingent state prices that are consistent with the given Arrow-Debreu (1952) state prices. These are derived by applying the forward equation to find the transition matrix that best fits the Arrow-Debreu (1952) state prices subject to the constraint that the resulting transition matrix has unimodal rows. The two top rows and two leftmost columns express the state variable in terms of both standard deviations from the current level and the stock price.

Panel A: State Prices on April 27, 2011												
Return\Tenor	0.25	0.5	0.75	1	1.25	1.5	1.75	2	2.25	2.5	2.75	3
−36%	0.005	0.023	0.038	0.050	0.058	0.064	0.068	0.071	0.073	0.075	0.076	0.076
−30%	0.007	0.019	0.026	0.030	0.032	0.034	0.034	0.035	0.035	0.035	0.034	0.034
−24%	0.018	0.041	0.046	0.050	0.051	0.052	0.051	0.050	0.050	0.049	0.048	0.046
−16%	0.045	0.064	0.073	0.073	0.072	0.070	0.068	0.066	0.064	0.061	0.058	0.056
−9%	0.164	0.156	0.142	0.128	0.118	0.109	0.102	0.096	0.091	0.085	0.081	0.076
0%	0.478	0.302	0.234	0.198	0.173	0.155	0.141	0.129	0.120	0.111	0.103	0.096
9%	0.276	0.316	0.278	0.245	0.219	0.198	0.180	0.164	0.151	0.140	0.130	0.120
20%	0.007	0.070	0.129	0.155	0.166	0.167	0.164	0.158	0.152	0.145	0.137	0.130
31%	0.000	0.002	0.016	0.036	0.055	0.072	0.085	0.094	0.100	0.103	0.105	0.105
43%	0.000	0.000	0.001	0.004	0.009	0.017	0.026	0.036	0.045	0.053	0.061	0.067
57%	0.000	0.000	0.000	0.000	0.000	0.000	0.001	0.001	0.002	0.002	0.003	0.003
Discount Factor	1.000	0.993	0.983	0.969	0.953	0.938	0.920	0.900	0.883	0.859	0.836	0.809

Panel B: The State Price Transition Matrix (P)												
	Sigmas	−5	−4	−3	−2	−1	0	1	2	3	4	5
Sigmas	$S_0 \backslash S_T$	0.64	0.70	0.76	0.84	0.91	1	1.09	1.20	1.31	1.43	1.57
−5	0.64	0.671	0.241	0.053	0.005	0.001	0.001	0.001	0.001	0.001	0.000	0.000
−4	0.70	0.280	0.396	0.245	0.054	0.004	0.000	0.000	0.000	0.000	0.000	0.000
−3	0.76	0.049	0.224	0.394	0.248	0.056	0.004	0.000	0.000	0.000	0.000	0.000

(Continued)

Table 2: Original Table II, part 1: state prices and first part of the transition price matrix.

Table II—Continued

Panel B: The State Price Transition Matrix (P)												
	Sigmas	−5	−4	−3	−2	−1	0	1	2	3	4	5
−2	0.84	0.006	0.044	0.218	0.390	0.250	0.057	0.003	0.000	0.000	0.000	0.000
−1	0.91	0.006	0.007	0.041	0.211	0.385	0.249	0.054	0.002	0.000	0.000	0.000
0	1.00	0.005	0.007	0.018	0.045	0.164	0.478	0.276	0.007	0.000	0.000	0.000
1	1.09	0.001	0.001	0.001	0.004	0.040	0.204	0.382	0.251	0.058	0.005	0.000
2	1.20	0.001	0.001	0.001	0.002	0.006	0.042	0.204	0.373	0.243	0.055	0.004
3	1.31	0.002	0.001	0.001	0.002	0.003	0.006	0.041	0.195	0.361	0.232	0.057
4	1.43	0.001	0.000	0.000	0.001	0.001	0.001	0.003	0.035	0.187	0.347	0.313
5	1.57	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.032	0.181	0.875

Table 3: Original Table II, part 2: continuation of the transition price matrix.

5.4 Table III: Recovered pricing kernel and natural transition matrix

Read:

- The natural transition matrix puts high probability mass near the diagonal, consistent with local movement in the stock index.
- The recovered kernel declines monotonically from 1.86 in the far-left state to 0.22 in the far-right state.
- The kernel equals 1 at the current state.
- Ross reports a quarterly characteristic root of 1.0018 in this implementation, emphasizing sensitivity in estimation.

Economic message: The table is the core empirical output. It separates option-implied state prices into a natural transition probability matrix and a pricing kernel.

Table III
The Recovered Pricing Kernel and the Natural Probability Transition Matrix

Applying the Recovery Theorem to the data in Table II, the matrix below displays the resulting natural transition probabilities from the ranges for the stock price returns in the far left column to the identical ranges in the top rows. The bottom row displays the recovered kernel for the given stock ranges in the top row.

The Natural Probability Transition Matrix (F)												
Sigmas		−5	−4	−3	−2	−1	0	1	2	3	4	5
Sigmas	$S_0 \backslash S_T$	−36%	−30%	−24%	−16%	−9%	0%	9%	20%	31%	43%	57%
−5	−36%	0.670	0.253	0.061	0.006	0.002	0.001	0.002	0.001	0.002	0.002	0.000
−4	−30%	0.266	0.395	0.267	0.066	0.005	0.000	0.001	0.000	0.000	0.000	0.000
−3	−24%	0.043	0.205	0.393	0.278	0.073	0.007	0.000	0.000	0.000	0.000	0.000
−2	−16%	0.004	0.035	0.193	0.390	0.290	0.081	0.006	0.000	0.000	0.000	0.000
−1	−9%	0.004	0.005	0.031	0.181	0.385	0.309	0.080	0.005	0.000	0.000	0.000
0	0%	0.003	0.004	0.011	0.031	0.132	0.477	0.333	0.010	0.000	0.000	0.000
1	9%	0.000	0.000	0.000	0.002	0.027	0.169	0.381	0.314	0.095	0.011	0.000
2	20%	0.000	0.000	0.000	0.001	0.003	0.028	0.163	0.373	0.318	0.102	0.013
3	31%	0.000	0.000	0.000	0.001	0.001	0.003	0.025	0.148	0.361	0.330	0.130
4	43%	0.000	0.000	0.000	0.000	0.000	0.000	0.001	0.019	0.131	0.347	0.501
5	57%	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.014	0.113	0.873
Kernel, $\phi =$		1.86	1.77	1.62	1.44	1.24	1	0.83	0.66	0.50	0.35	0.22

Table 4: Original Table III: recovered kernel and natural transition matrix.

5.5 Table IV: Recovered and bootstrapped natural marginal distributions

Read:

- Panel A gives recovered marginal distributions across horizons from 0.25 to 3 years.
- Panel B reports recovered annualized means, volatilities, risk-free rates, risk premia, Sharpe ratios, and at-the-money implied volatilities.
- Recovered expected returns are around 5–6% per year, below the historical mean of about 10.3%.
- Recovered volatility is around 15%, close to the historical annualized volatility of 15.5%.
- The recovered Sharpe ratio declines with horizon and turns negative by 2.5 years in this implementation.

Economic message: The recovered distribution is more conservative on expected return but still captures volatility. The risk premium estimates are sensitive to the risk-free-rate estimates embedded in state prices.

Table IV
The Recovered and the Bootstrapped Natural Marginal Distributions

Each column of Panel A represents a time horizon (years) and the entries in that column are the recovered probabilities of the respective future S&P 500 ranges. Panel B computes the associated summary statistics for the recovered marginal distribution of each time horizon. Panel C displays the comparable summary statistics derived from monthly S&P 500 returns over the 50-year period from 1960 to 2010.

Panel A: The Recovered Marginal Distributions												
Return\Tenor	0.25	0.5	0.75	1	1.25	1.5	1.75	2	2.25	2.5	2.75	3
−36%	0.003	0.012	0.020	0.026	0.030	0.032	0.034	0.036	0.037	0.038	0.038	0.039
−30%	0.004	0.010	0.014	0.016	0.017	0.018	0.018	0.018	0.018	0.018	0.018	0.018
−24%	0.011	0.025	0.028	0.030	0.030	0.030	0.030	0.029	0.029	0.028	0.028	0.027
−16%	0.031	0.044	0.049	0.049	0.047	0.046	0.044	0.043	0.041	0.039	0.038	0.038
−9%	0.132	0.124	0.111	0.099	0.090	0.083	0.077	0.072	0.068	0.065	0.062	0.059
0%	0.477	0.299	0.228	0.190	0.165	0.146	0.132	0.121	0.112	0.104	0.098	0.092
9%	0.333	0.377	0.327	0.285	0.252	0.225	0.203	0.185	0.171	0.159	0.148	0.140
20%	0.010	0.105	0.190	0.226	0.239	0.238	0.232	0.224	0.215	0.206	0.197	0.189
31%	0.000	0.005	0.031	0.068	0.104	0.134	0.157	0.173	0.184	0.192	0.197	0.200
43%	0.000	0.000	0.002	0.010	0.025	0.045	0.070	0.094	0.118	0.141	0.163	0.182
57%	0.000	0.000	0.000	0.000	0.001	0.002	0.003	0.005	0.007	0.009	0.012	0.015
Panel B: Recovered Summary Statistics (Annualized)												
	0.25	0.5	0.75	1	1.25	1.5	1.75	2	2.25	2.5	2.75	3
Mean	0.051	0.055	0.06	0.062	0.063	0.063	0.063	0.062	0.061	0.06	0.058	0.057
Sigma	0.117	0.14	0.147	0.15	0.152	0.153	0.154	0.154	0.153	0.152	0.151	0.149
Risk-free	0.005	0.002	0.01	0.018	0.026	0.034	0.041	0.048	0.054	0.061	0.068	0.074
$E - r$	0.047	0.053	0.05	0.043	0.036	0.03	0.022	0.015	0.007	−0.001	−0.009	−0.017
Sharpe	0.399	0.376	0.34	0.287	0.239	0.193	0.146	0.096	0.044	−0.008	−0.061	−0.115
ATM vol.	0.145	0.167	0.177	0.182	0.185	0.188	0.191	0.193	0.196	0.198	0.201	0.203

(Continued)

Table 5: Original Table IV, part 1: recovered marginal distributions and annualized summary statistics.

5.6 Table V: Recovered versus historical tail probabilities

Read:

- At the six-month horizon, the bootstrapped probability of a decline beyond 32% is 0.0008; the recovered probability is 0.0120.
- The bootstrapped probability of a decline beyond 26% is 0.0020; the recovered probability is 0.0223.
- The recovered distribution puts substantially more mass in the far-left tail.
- The historical distribution puts relatively more mass around moderate positive returns.

Economic message: This is the paper’s “dark matter” point. Option prices imply that catastrophic downside states are more likely than raw historical frequency would suggest, unless the entire difference is attributed to risk premia.

Table IV—Continued

Panel C: Annualized Monthly S&P 500 Returns	
Mean	0.103
Sigma	0.155
Risk-free	0.055
$E - r$	0.049
Sharpe	0.316

Table V

**The Densities and the Cumulative Distributions for the Recovered
and the Bootstrapped Natural Probabilities**

(Six month horizon, bootstrap using data from 1/1/1960 to 11/30/2010)

The rows of the table correspond to ranges for the S&P 500 index for six months from the date April 27, 2011. The first and third columns are from the historical distribution obtained by bootstrapping independent monthly return observations from the period 1960 to 2010. The second and fourth columns display the comparable distribution results from the recovered distribution of Table IV.

Range	Densities		Distribution Functions	
	Bootstrapped	Recovered	Bootstrapped	Recovered
−32%	0.0008	0.0120	0.0008	0.0120
−26%	0.0012	0.0103	0.0020	0.0223
−19%	0.0102	0.0250	0.0122	0.0473
−12%	0.0448	0.0438	0.0570	0.0912
−4%	0.1294	0.1242	0.1864	0.2153
0%	0.2834	0.2986	0.4698	0.5139
4%	0.3264	0.3765	0.7962	0.8904
14%	0.1616	0.1047	0.9578	0.9951
24%	0.0384	0.0047	0.9962	0.9998
35%	0.0036	0.0002	0.9998	1.0000
48%	0.0002	0.0000	1.0000	1.0000

Table 6: Original Table IV continuation and Table V: recovered versus bootstrapped natural probabilities.

6 Short summary

- Option prices reveal state prices, not natural probabilities directly.
- State prices combine beliefs and risk aversion through the pricing kernel.
- Ross asks whether the natural distribution and pricing kernel can be recovered from state prices alone.
- The answer is yes under no arbitrage, irreducibility, and transition-independent pricing kernels.
- In finite state space, the key equation is $DP = \delta FD$.
- Because rows of F sum to one, recovery becomes the eigenvalue problem $Pz = \delta z$.
- Perron-Frobenius gives a unique positive eigenvalue and eigenvector for an irreducible nonnegative pricing matrix.
- The eigenvalue gives the subjective discount factor and the eigenvector gives the pricing kernel.
- The natural transition matrix is then $F = (1/\delta)DPD^{-1}$.
- The theorem does not require historical returns or a parametric utility function.
- The multinomial recovery theorem clarifies that full rank is important for uniqueness.
- The empirical illustration estimates state prices from S&P 500 option prices and applies recovery.
- The recovered pricing kernel falls as the stock state rises.
- The recovered natural distribution has a fatter left tail than the bootstrapped historical distribution.
- The recovered six-month probability of a crash beyond 26% is much larger than the historical bootstrap probability.
- The recovered pricing kernel also produces bounds on Sharpe ratios and return-predictability R^2 values.
- Ross stresses that empirical recovery is sensitive and requires better state-price transition estimation.

7 Conclusion

7.1 Core conclusion

Ross's Recovery Theorem shows that, under strong but economically interpretable conditions, state prices contain enough information to recover both the pricing kernel and the market's natural probability distribution. The theorem turns a seemingly underidentified problem into an eigenvalue problem.

7.2 Economic intuition

Risk-neutral probabilities are distorted by risk aversion. Recovery works because these distortions must be dynamically consistent across states and horizons. A transition-independent kernel imposes exactly the separability needed to undo the distortion.

7.3 Takeaway

The paper provides a theoretical bridge from option markets to market beliefs. It also provides a new way to think about crash probabilities, risk premia, and efficient-market tests. Its empirical estimates are preliminary, but the conceptual message is powerful: derivative prices may contain much more information about beliefs and risk aversion than risk-neutral probabilities alone reveal.